

50 Python For-Loop Practice Questions

Instructions: Solve each problem using only for loops. Do NOT use function definitions or dictionaries. Lists may be used where explicitly given in a question.

Easy (20 questions)

1. Print numbers from 1 to 10 using a for loop.
2. Print all even numbers between 2 and 20 using a for loop.
3. Print all odd numbers between 1 and 19 using a for loop.
4. Print the multiplication table of 5 (from 1 to 10) using a for loop.
5. Calculate and print the sum of numbers from 1 to 50 using a for loop.
6. Print each character of the string 'python' on a new line using a for loop.
7. Given the list numbers = [2, 4, 6, 8, 10], print each element multiplied by 3 using a for loop.
8. Given the string s = 'hello', print the index and character for each character using a for loop (format: index: character).
9. Use a for loop to print the squares of numbers from 1 to 10.
10. Given the list nums = [1, 3, 5, 7, 9], print the cumulative sum at each step using a for loop.
11. Print the first 10 multiples of 7 using a for loop.
12. Given text = 'abracadabra', count how many times the letter 'a' appears using a for loop.
13. Use a for loop to print numbers from 10 down to 1.
14. Given the list values = [10, 20, 30, 40], print each value divided by 10 using a for loop.
15. Print all lowercase letters in the string 'AbCdEFG' using a for loop.
16. Given the list items = ['apple', 'banana', 'cherry'], print each item in uppercase using a for loop.
17. Use a for loop to print the first 15 Fibonacci numbers starting with 0 and 1 (do it inline, without defining functions).
18. Given the list mixed = [1, '2', 3, '4', 5], print only the integer elements using a for loop.
19. Use a for loop to print numbers from 1 to 30 that are divisible by both 2 and 3.
20. Print all characters of the string 'Data' reversed using a for loop (do not use reversed() or slicing).

Intermediate (25 questions)

21. Given the list nums = [12, 5, 8, 130, 44], find and print the largest number using a for loop.
22. Given a string sentence = 'this is a test', count and print the number of words using a for loop (assume words are separated by single spaces).
23. Given list nums = [3, 1, 4, 1, 5, 9, 2], print the list of numbers that are greater than their immediate predecessor using a for loop.

24. 24. Given the list numbers = [2, 4, 6, 8, 10], create and print a new list containing only the odd numbers using a for loop.
25. 25. Given text = 'Level', check (using a for loop and case normalization) whether the text is a palindrome and print 'Palindrome' or 'Not palindrome'.
26. 26. Given the list primes = [2,3,5,7,11,13,17], print all primes that are two-digit numbers using a for loop.
27. 27. Using a for loop, compute and print the factorial of 8 (do not use math.factorial or functions).
28. 28. Given list nums = [2, -3, 5, -1, 0, -7], count and print how many negative numbers are in the list using a for loop.
29. 29. Given the list words = ['cat', 'dog', 'horse', 'cow'], print only those words with length greater than 3 using a for loop.
30. 30. Given string s = 'Python3.10', extract and print all digits in the string using a for loop.
31. 31. For numbers 1 to 50, print 'Fizz' for multiples of 3, 'Buzz' for multiples of 5, and the number itself otherwise (use a for loop).
32. 32. Given nums = [10, 20, 30, 40, 50], compute and print the average using a for loop.
33. 33. Given a list of strings names = ['anna', 'Bob', 'ALICE', 'Tim'], print them with the first letter capitalized using a for loop.
34. 34. Given list nums = [5, 2, 9, 1, 5, 6], use a for loop to create and print a sorted version of the list using simple selection sort logic (implement selection sort using for loops, no built-in sort()).
35. 35. Given text = 'a b c d e', replace every space with '-' and print the new string using a for loop.
36. 36. Given the list seq = [1,2,3,4,5], print all possible pairs (i, j) where i comes before j using nested for loops.
37. 37. Given the string s = 'aaabbbccc', compress consecutive characters and print the result as 'a3b3c3' using a for loop.
38. 38. Given numbers = [4, 9, 16, 25], print a new list of square roots (as integers) using a for loop.
39. 39. Given list words = ['red', 'blue', 'green', 'red', 'blue'], print each unique word only once using a for loop (do not use set()).
40. 40. Given a positive integer n = 28, use a for loop to find and print all its proper divisors (excluding n) and state whether n is perfect (sum of proper divisors equals n).
41. 41. Given the string s = 'MixEdCase', count uppercase and lowercase letters separately using a for loop and print both counts.
42. 42. Given list nums = [1,2,3,4,5,6,7,8,9], use a for loop to print a 3x3 matrix (rows) representation of the list (three numbers per row).
43. 43. Given list nums = [0,1,2,3,4,5], shift all elements to the right by one position using a for loop (the last element becomes first).
44. 44. Given text = 'abracadabra', print the index positions of each occurrence of the letter 'a' using a for loop.

45. 45. Given list numbers = [11,22,33,44,55], create and print a new list where each element is the original element minus the previous element (for the first element, subtract 0).

Hard (5 questions)

46. 46. Given two lists of possibly different lengths, a = [1,3,5,7] and b = [2,4,6], merge them into a single list by alternating elements (start with a[0]) using only for loops and indexing. If one list runs out, append the remaining elements of the longer list.
47. 47. Given a 4x4 matrix represented as a flat list m = [1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16], use nested for loops to rotate the matrix 90 degrees clockwise and print the rotated matrix as rows.
48. 48. Given a list of integers nums, use for loops to find the longest contiguous subsequence whose elements form an increasing sequence and print its length and elements. (Implement scanning logic with for loops and indexing.)
49. 49. Given list nums = [2,3,5,7,11,13,17,19], using only for loops, find and print all pairs (i, j) where $i < j$ and $\text{nums}[i] + \text{nums}[j]$ is a prime number.
50. 50. Given a string of digits s = '121332441', using for loops only, find and print the length of the longest palindrome substring that is centered at any single character (odd-length palindromes).